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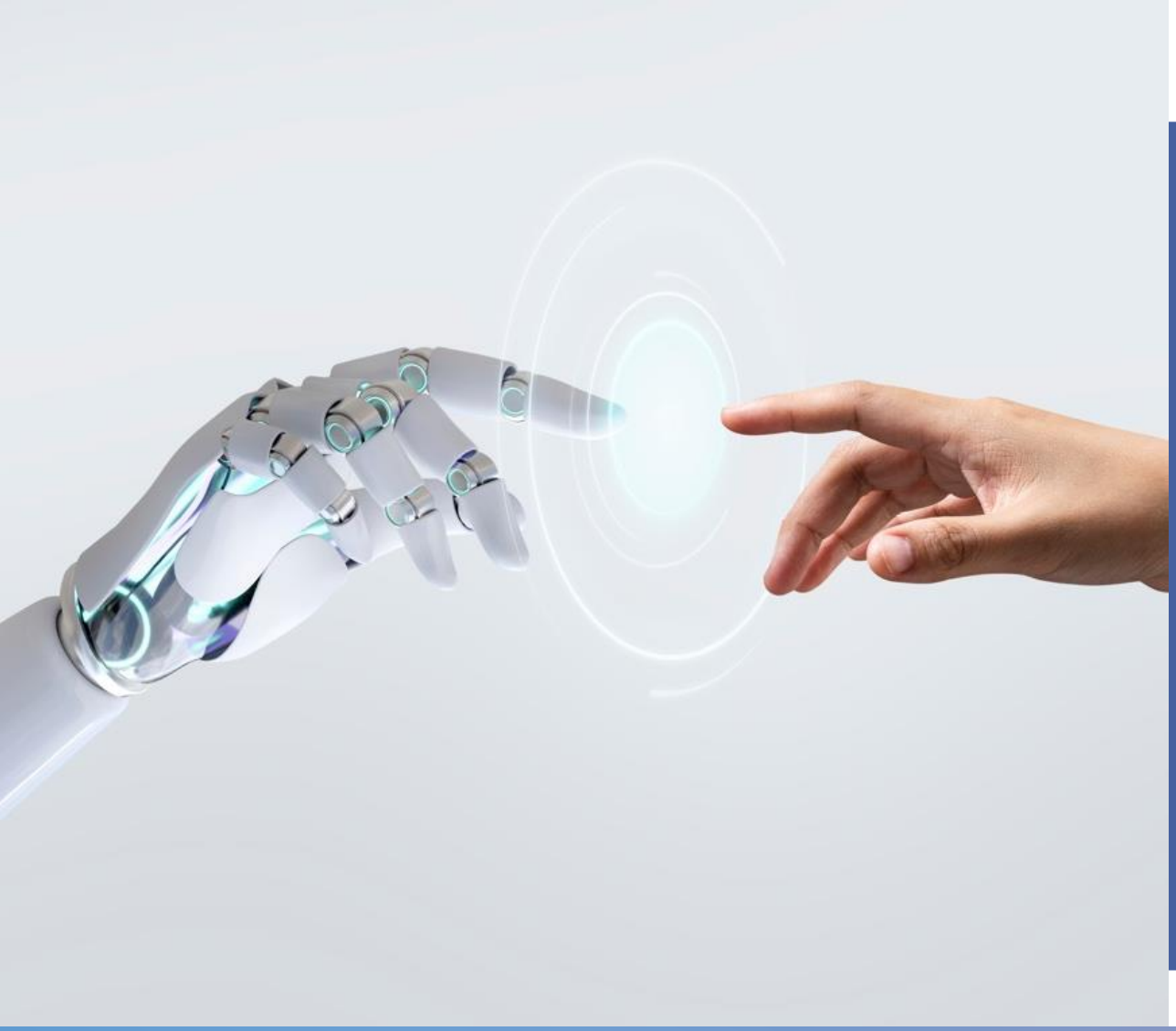
systems

CLOUD DATA ENGINEERING (AZURE)

By: Ahmed Irshad Hussain (35820)
KSA Data Analytics

www.systemsltd.com



Azure Data Engineering End to End Project

Azure Data Engineering

Step 1:

Create Resource Group on Microsoft Azure.

(Download a data source first, I download Olympics Data Source from Kaggle.)

Step 2:

Now create Azure Data Lake Storage.

Go to Storage Accounts, fill the details then review + create.

Data Lake Storage

Hierarchical namespace	Enabled
Default access tier	Hot
Blob anonymous access	Disabled

Hierarchal namespace should be enabled. Data Lake Gen2 must be upgraded for it!

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Basics Tags Review + create

Automation Link

Basics

Subscription	Azure subscription 1
Resource group name	Ahmed-retaildata-rg
Region	East US

Tags

None

Create Resource Groups on Azure using your account.

Create a storage account

Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.

Subscription * Azure subscription 1

Resource group * Ahmed-retail-data-rg
Create new

Instance details

Storage account name * ahmedretaildatalake123

Region * (US) East US
Deploy to an Azure Extended Zone

Preferred storage type Choose preferred storage type

This helps us provide relevant guidance. It doesn't restrict your storage to this resource type. Learn more

Performance *
- **Standard:** Recommended for most scenarios (general-purpose v2 account)
- **Premium:** Recommended for scenarios that require low latency.

Redundancy * Geo-redundant storage (GRS)

- [x] Make read access to data available in the event of regional unavailability.
- [ ] Geo priority replication guarantees Blob storage data is geo-replicated within 15 minutes.

Name your storage account, the rest should the same as of resource group

Previous Next Review + create

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Home

ahmedretaildatalake123_1773050527132 | Overview

Deployment

Search

- Overview
- Inputs
- Outputs
- Template

Delete | Cancel | Redeploy | Download | Refresh

Your deployment is complete

Deployment name : ahmedretaildatalake123_1773050527132
Subscription : Azure subscription 1
Resource group : Ahmed-retail-data-rg

Start time : 3/9/2026, 3:02:56 PM
Correlation ID : fc0108d0-ca50-4705-8107-d4dc8556ae14

Deployment details

Resource	Type	Status	Operation details
ahmedretaildatalake123/default	Microsoft.Storage/storageAccou	OK	Operation details
ahmedretaildatalake123/default	Microsoft.Storage/storageAccou	OK	Operation details
ahmedretaildatalake123	Storage account	OK	Operation details

Next steps

Go to resource

Azure Data Engineering

Step 3:

Create Architectural Folders.

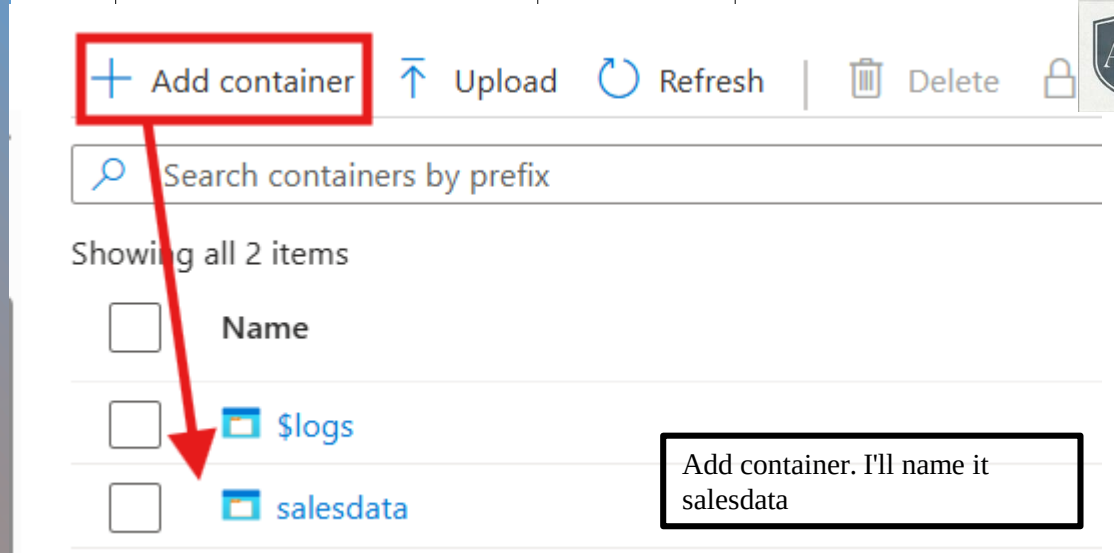
Go to storage account and create containers.

1. Create salesdata container.
2. Open sales data container and add 3 directories.

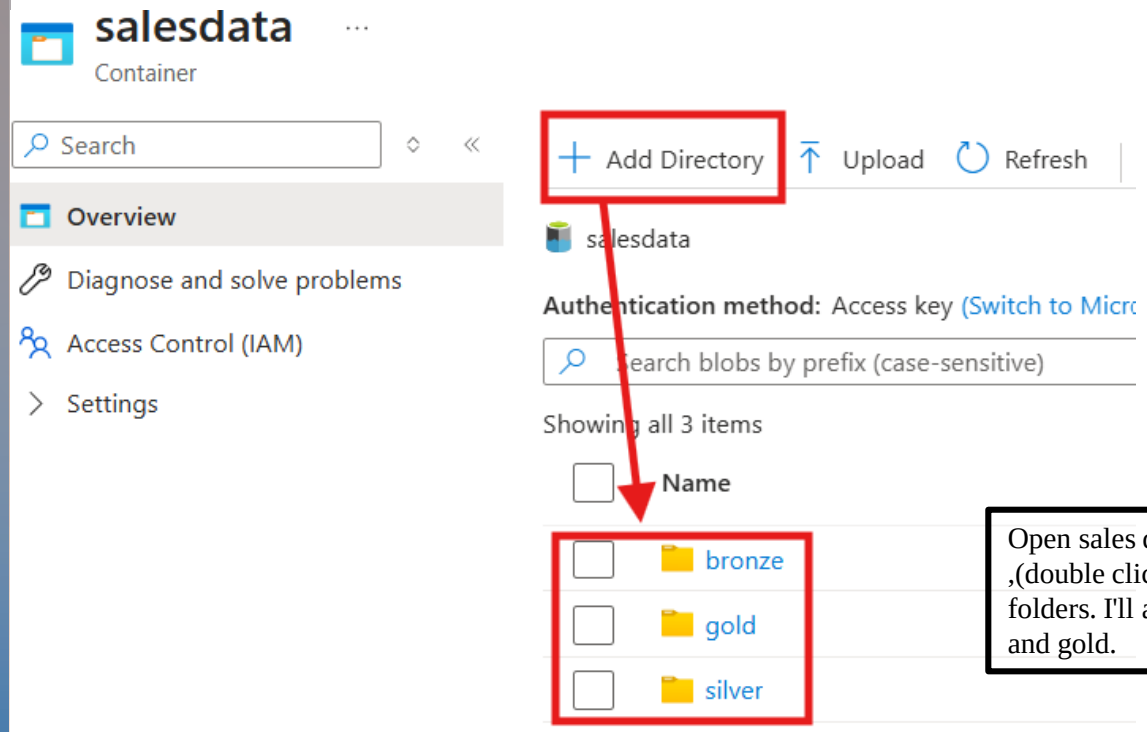
   Bronze: For raw data.

   Silver: For cleaned data.

   Gold: For aggregated business data.



Add container. I'll name it salesdata



Open sales data container ,(double click), and add 3 folders. I'll add bronze, silver and gold.

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Azure Data Engineering

Step 4:

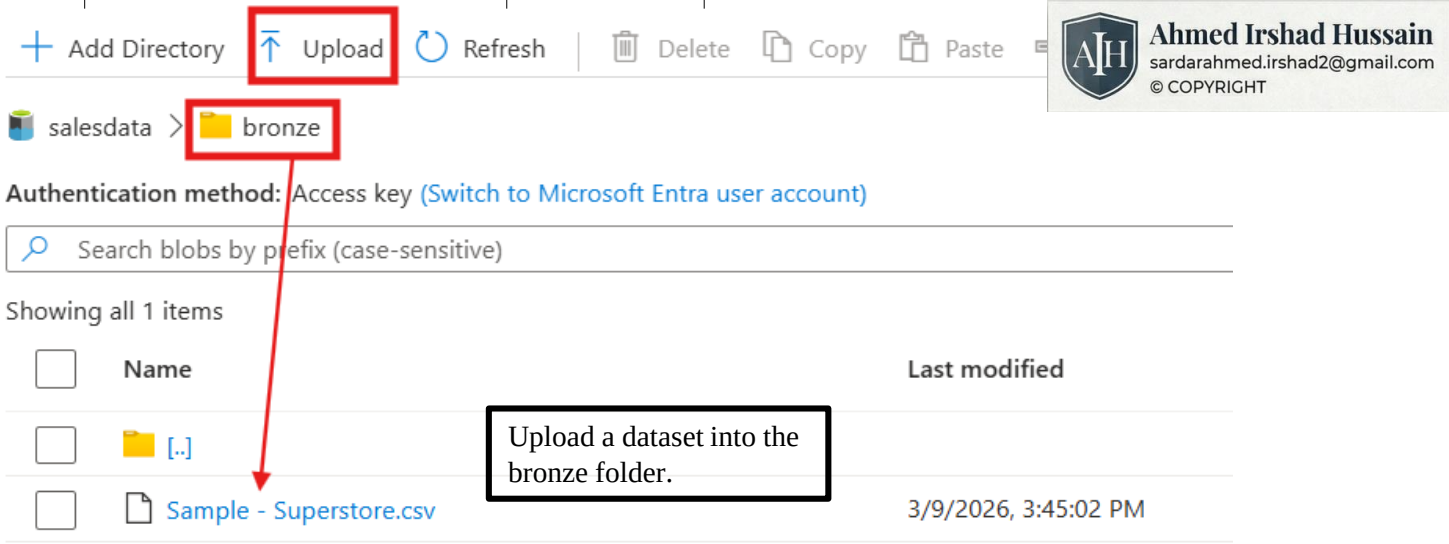
Upload dataset to Bronze Layer.

Download a sample data set from google.

I download a superstore sales dataset from Kaggle.

Step 5:

Create Azure Databricks (Spark Environment).



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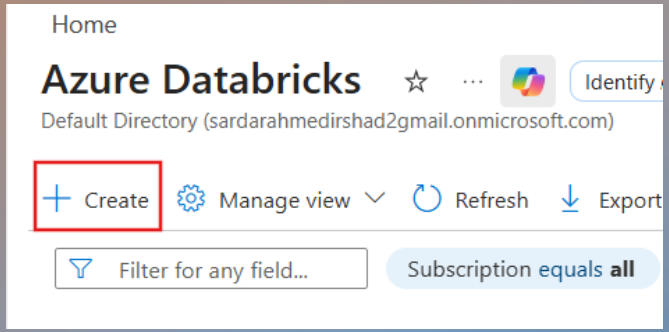
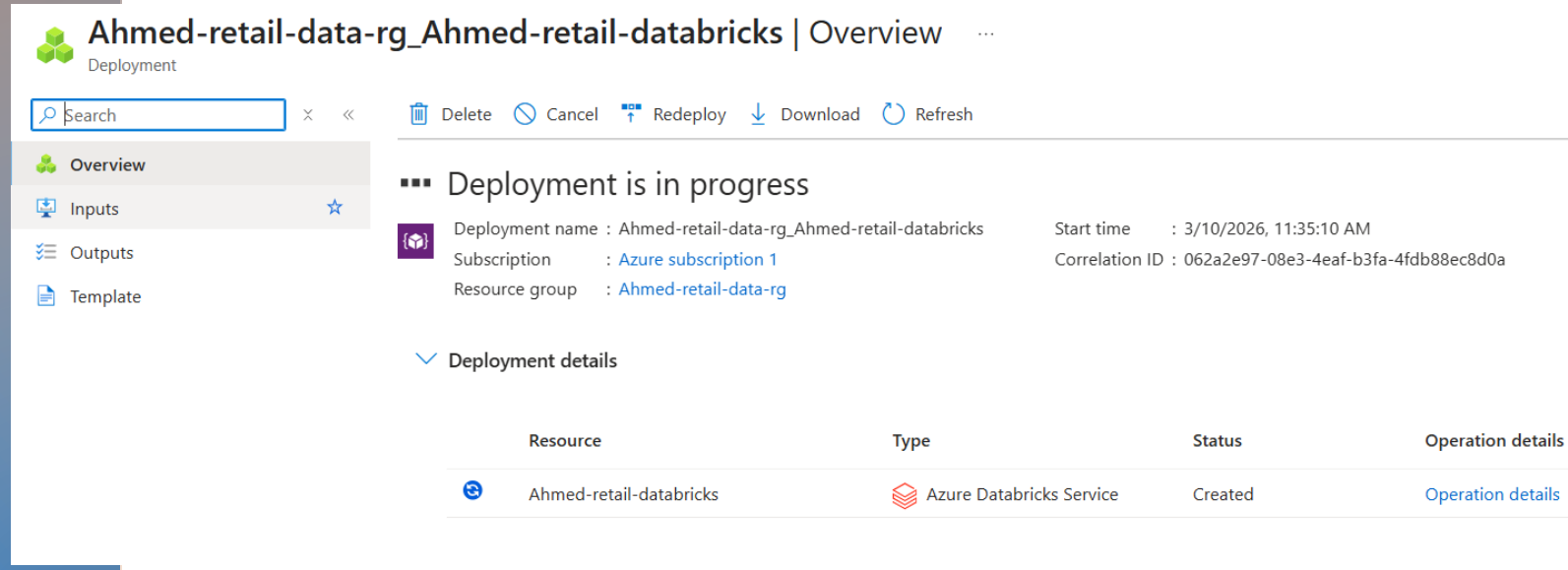


1. Search azure databricks.
2. Click create.
3. Fill.
4. Launch Databricks.



Azure Data Engineering

Step 6:

Create Spark Cluster.

Inside Databricks, go to compute.

Step 7:

Create Notebook (Spark Code).

Step 8:

Connect Databricks to Data Lake.

Mount the storage.

```
Just now (1s)                                                    1

storage_account = "ahmedretaildatalake123"
container = "salesdata"

data_path = f"abfss://{container}@{storage_account}.dfs.core.windows.net/bronze"
```

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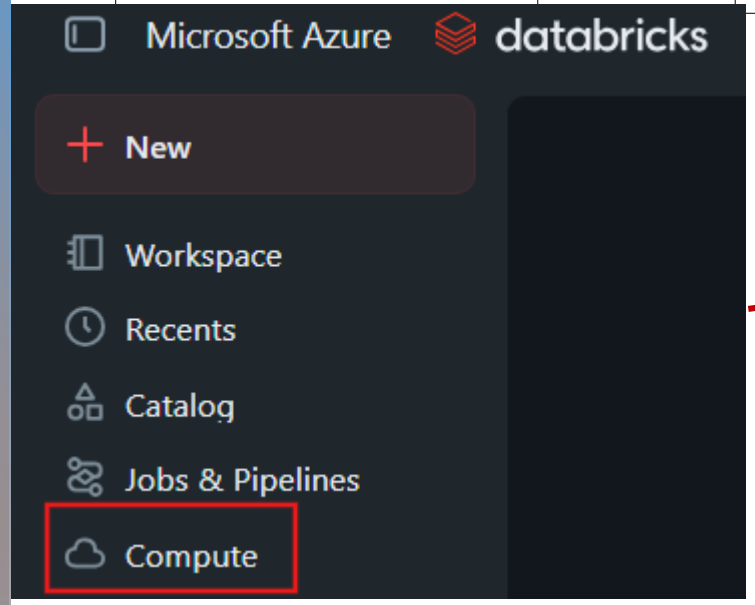
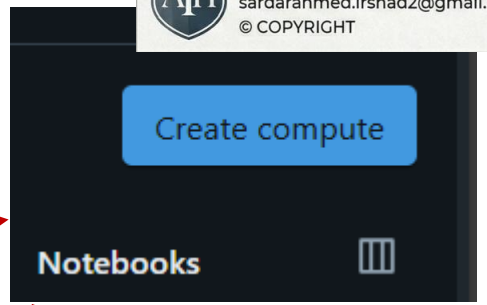
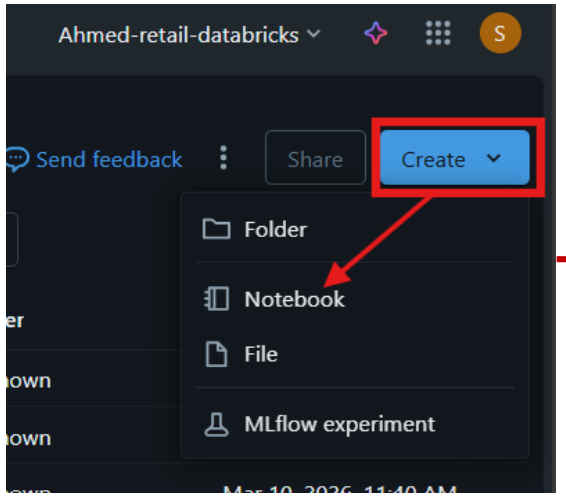
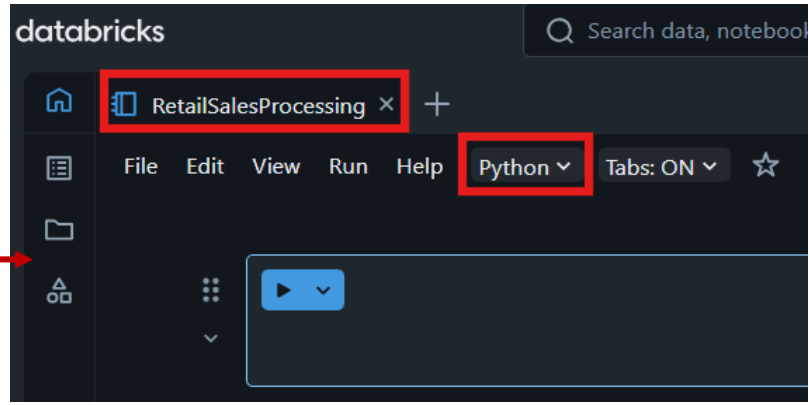




1. Inside Databricks, go to Compute.
2. Create New Compute
3. Fill.

State	Name	Runtime	Active memory	Active cores	Active DBU / h	Source	Creator	Notebooks
●	sardarahmed.irshad2@gmail.com's Cluster 202...	17.3	16 GB	4 cores	2	UI	sardarahmed.irshad2	-





1. Inside Databricks, go to workspace.
2. Create a new notebook.
3. Name it & select Python.

Azure Data Engineering

Step 9:

Give Databricks permission to access your Azure Data Lake Storage.

1. Get Storage Account Access Key.
2. Configure Databricks to Access Data Lake.

Step 10:

Read Bronze Data (Raw Data).

Load the CSV file.

Just now (10s) 3

```
data_path = "abfss://salesdata@ahmedretaildatalake123.dfs.core.windows.net/bronze/Sample - Superstore.csv"

df = spark.read.format("csv") \
.option("header","true") \
.option("inferSchema","true") \
.load(data_path)

df.show()
```

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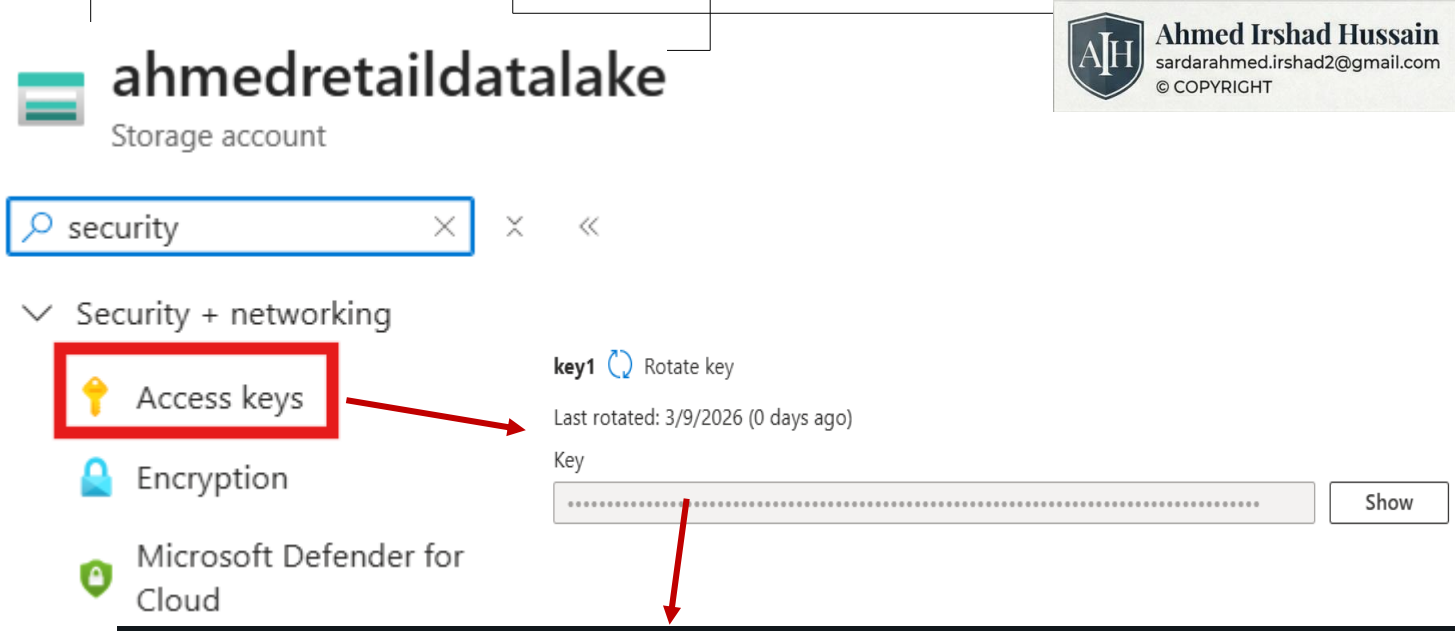

12:25 PM (<1s) 2

```
spark.conf.set(
"fs.azure.account.key.ahmedretaildatalake123.dfs.core.windows.net",
"[redacted]DsdbgWatq3v6+D/A+1+AStBMRF/w=="
)
```

1. Search for "Security + Networking"
2. Go to Access Keys and copy the code.
3. Paste the code in the python script.
4. Run the code.

Azure Data Engineering

Step 11 (Silver Layer):

Clean the data.

1. Remove Null Values.
2. Remove Duplicates.

Step 12:

Saving the clean data to silver folder which I created earlier.

```
Just now (1s)                                                    12
df_clean.write.mode("overwrite") \
.parquet("abfss://salesdata@ahmedretaildatalake123.dfs.core.windows.net/silver/")
(1) Spark Jobs
```

```
Just now (<1s)
df_clean = df.dropna()
df_clean: pyspark.sql.dataframe.DataF
```

```
Just now (<1s)
df_clean = df_clean.dropDuplicates()
df_clean: pyspark.sql.dataframe.DataFrame = [
```

```
Just now (<1s)
df_clean = df_clean.withColumn(
    "Revenue",
    col("Sales") - col("Discount")
)
df_clean: pyspark.sql.dataframe.DataFrame
```

1. First remove null values.
2. Remove duplicates.
3. Then I created a revenue column.

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Azure Data Engineering

Step 13 (Gold Layer):

Aggregated Data.

Create Business metrics.

Save the aggregated data to gold folder.

Now I have,
Bronze → Raw
Silver → Cleaned
Gold → Business analytics

Just now (1s) 14

```
df_gold = df_clean.groupBy("Region") \
.sum("Revenue")
df_gold.write.mode("overwrite") \
.parquet("abfss://salesdata@ahmedretaildatalake123.dfs.core.windows.net/gold/")
```

(2) Spark Jobs

df_gold: pyspark.sql.dataframe.DataFrame = [Region: string, sum(Revenue): double]

1. First, I created the metric, total revenue by regions.
2. Then save it to gold folder.

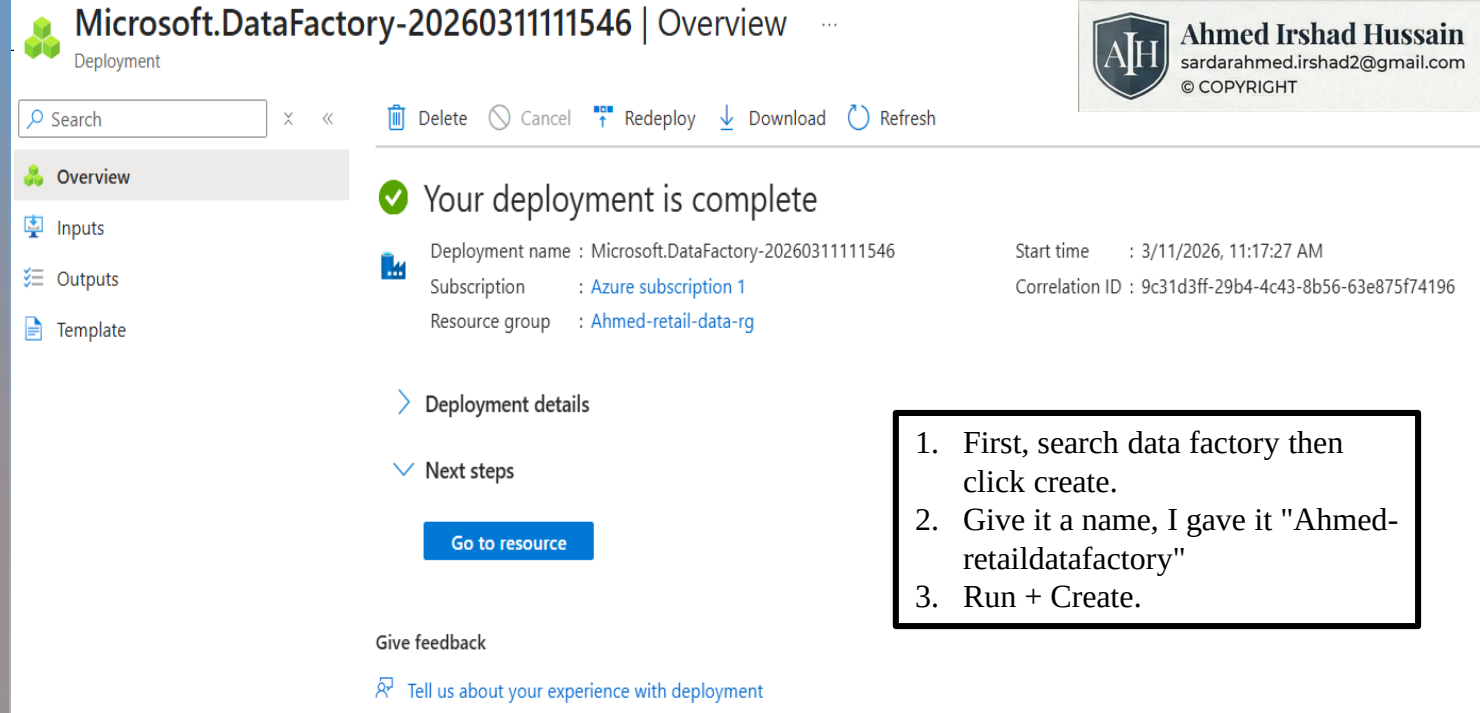
Azure Data Engineering

Step 14 :

Create Azure Data Factory (Pipeline).

Search "Azure Data Factory", then click create.

Save the aggregated data to gold folder.



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1. First, search data factory then click create.
2. Give it a name, I gave it "Ahmed-retaildatafactory"
3. Run + Create.

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New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more

Select a data store

Search

All | Azure | Database | File | Generic protocol

Azure AI Search
Azure Blob Storage
Azure Cosmos DB for MongoDB
Azure Cosmos DB for NoSQL
Azure Data Explorer (Kusto)
Azure Data Lake Storage Gen2
Azure Database for MySQL
Azure Database for PostgreSQL
Azure Databricks Delta Lake

Continue | Cancel

Select format

Choose the format type of your data

Avro
Binary
DelimitedText
Excel
Iceberg
JSON
ORC
Parquet
XML

Continue | Back | Cancel

Set properties

Name

DelimitedText1

Linked service *

Select...

Filter...

+ New

1. First, search data factory then click create.
2. Give it a name, I gave it "Ahmed-retaildatafactory"
3. Run + Create.

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New linked service

Azure Data Lake Storage Gen2 Learn more

Name *
Ahmed_RetailDataLakeStorage

Description

Connect via integration runtime *
AutoResolveIntegrationRuntime

Authentication type
Account key

Account selection method
From Azure subscription Enter manually

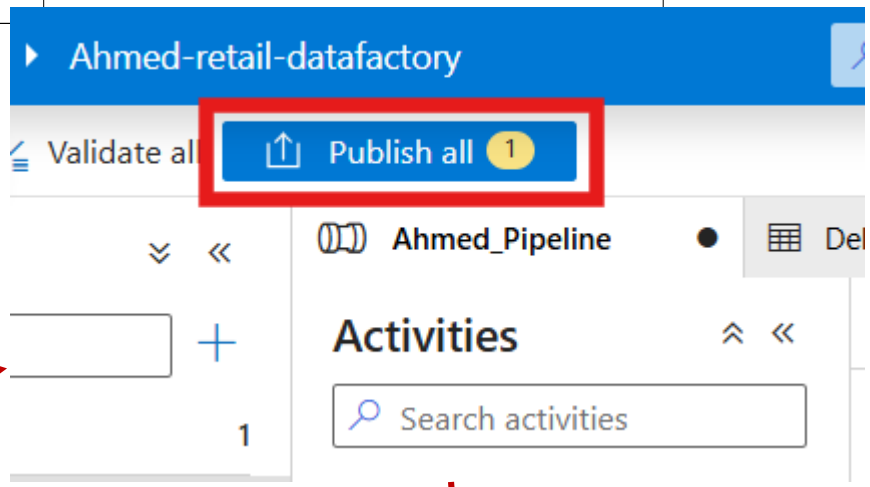
Azure subscription
Select all

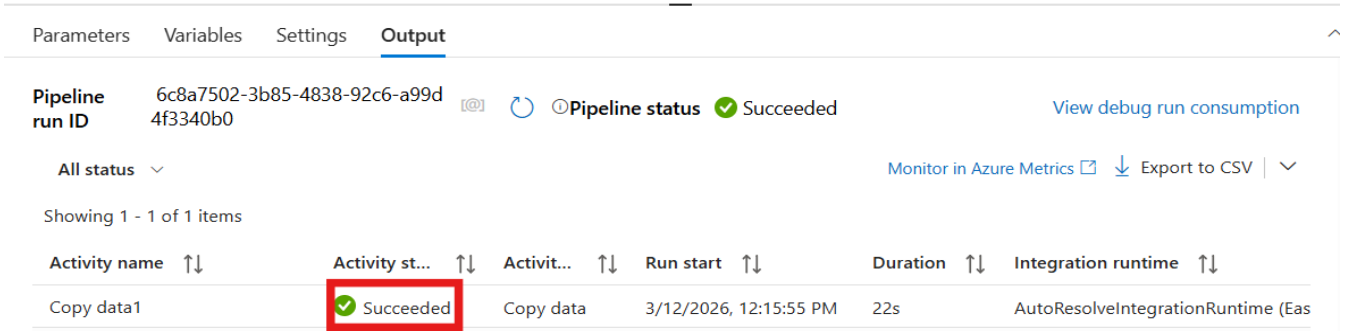
Storage account name *
ahmedretaildatalake123

Test connection
To linked service To file path

Connection successful
Test connection

Create Cancel





1. Link the service, add credentials. And create, the connection should be successful.
2. After that click "Copy Data1" and go to mapping and import schemas
3. Publish!
4. Debug and status of ADF should be succeeded.

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Add Directory | Upload | Refresh | Delete | Copy | Paste | Rename | Acquire lease

salesdata > silver

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 7 items

Name	Last modified	Access tier
[..]		
_SUCCESS	3/10/2026, 1:57:34 PM	Hot (Inferred)
_committed_8887116624557104037	3/10/2026, 1:57:34 PM	Hot (Inferred)
_started_8887116624557104037	3/10/2026, 1:57:34 PM	Hot (Inferred)
part-00000-tid-8887116624557104037-1621f001-1218-490d...	3/10/2026, 1:57:34 PM	Hot (Inferred)
part-00001-tid-8887116624557104037-1621f001-1218-490d...	3/10/2026, 1:57:34 PM	Hot (Inferred)
part-00002-tid-8887116624557104037-1621f001-1218-490d...	3/10/2026, 1:57:34 PM	Hot (Inferred)
superstore.parquet	3/12/2026, 12:16:13 PM	Hot (Inferred)

ate the current resource | Sink | Mapping | Settings | User properties

Source	Type	Destination	Type
Row ID	abc String	row_id	UTF8
Order ID	abc String	order_id	UTF8
Order Date	abc String	order_date	UTF8
Ship Date	abc String	ship_date	UTF8
Ship Mode	abc String	ship_mode	UTF8

2 minutes ago (2s) 3 Python

```
df = spark.read.parquet(
"abfss://salesdata@ahmedretaildatalake123.dfs.core.windows.net/silver/superstore.parquet"
)

df.display()
```

(2) Spark Jobs

df: pyspark.sql.dataframe.DataFrame = [row_id: string, order_id: string ... 19 more fields]

Table

	row_id	order_id	order_date	ship_date	ship_mode	customer_id	customer_name	segment	country
1	1	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
2	2	CA-2016-152156	11/8/2016	11/11/2016	Second Class	CG-12520	Claire Gute	Consumer	United States
3	3	CA-2016-138688	6/12/2016	6/16/2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States
4	4	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
5	5	US-2015-108966	10/11/2015	10/18/2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States
6	6	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
7	7	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
8	8	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
9	9	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
10	10	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
11	11	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
12	12	CA-2014-115812	6/9/2014	6/14/2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States
13	13	CA-2017-114412	4/15/2017	4/20/2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States
14	14	CA-2016-161389	12/5/2016	12/10/2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States
15									

Verifying!

In ADF, the destination columns and in Databricks are the same

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Azure Data Engineering Pipeline

Bronze Layer (CSV)

Azure Data Lake Storage

salesdata/bronze/

superstore.csv

Azure Data Factory (ADF)

Pipeline: Ahmed_Pipeline

Copy Data Activity

- Source: DelimitedText1 (CSV)
- Sink: SilverSuperstoreDataset (Parquet)

Data Transferred Successfully

Silver Layer (Parquet)

Azure Data Lake Storage

salesdata/silver/

superstore.parquet
(Clean + Optimized)

Databricks / Apache Spark

Read Parquet Data

- spark.read.parquet()
- salesdata/silver/

Data Cleaning

- Handle Nulls
- Remove Duplicates
- Transform Columns

Data Analysis

- Aggregations
- Filters & Joins
- Business Insights

Visualization / Export

- Power BI Reports
- SQL Analytics
- Dashboards

Azure Data Lake · Azure Data Factory · Azure Databricks · Apache Spark · Power BI